



Masterarbeit

ScholarCopilot with Fine-Grained Citation Retrieval

Eberhard Karls Universität Tübingen
Mathematisch-Naturwissenschaftliche Fakultät
Wilhelm-Schickard-Institut für Informatik
Lernbasierte Computer Vision
Niklas Munkes, niklas.munkes@student.uni-tuebingen.de, 2025

Bearbeitungszeitraum: 14.04.-14.10.2025

Gutachter:	Prof. Dr. Andreas Geiger, Universität Tübingen
Zweitgutachter:	Prof. Dr. Carsten Eickhoff, Universität Tübingen
Betreuer:	Haoyu He, Universität Tübingen

Selbstständigkeitserklärung

Hiermit versichere ich, dass ich die vorliegende Masterarbeit selbständig und nur mit den angegebenen Hilfsmitteln angefertigt habe und dass alle Stellen, die dem Wortlaut oder dem Sinne nach anderen Werken entnommen sind, durch Angaben von Quellen als Entlehnung kenntlich gemacht worden sind. Diese Masterarbeit wurde in gleicher oder ähnlicher Form in keinem anderen Studiengang als Prüfungsleistung vorgelegt.

Niklas Munkes (Matrikelnummer 4269436), August 21, 2025

Abstract

TODO

Contents

1	Introduction	9
2	Related Work	11
2.1	Decoder Architectures	11
2.2	Retrieval-Augmented Generation	11
3	Methods	13
3.1	ScholarCopilot	13
4	Experiments	15
5	Conclusion	17

1 Introduction

TODO

2 Related Work

TODO

2.1 Decoder Architectures

TODO

2.2 Retrieval-Augmented Generation

TODO

Previous work on text generation with citations treated retrieval and generation as separate stages during inference. For example, "Attribute First, then Generate" [SHC⁺24] initially retrieves a set of relevant spans which are then grouped into coherent clusters which in turn are used to inform an iterative sentence-by-sentence generation step [SHC⁺24]. Similarly, OpenScholar [AHS⁺24] first retrieves and reranks a set of papers relevant to the given query. These papers inform a subsequent generation step that produces a generated response r_i [AHS⁺24]. This response is refined through iterative *self-feedback* where during each iteration of the feedback loop $F = f_1, \dots, f_T$, the generator model generates sentences f_j that describe potential improvements to the response and may also prompt the retrieval pipeline to retrieve additional papers, should r_i require more information to generate an improved response r_{i+1} [AHS⁺24]. Depending on the choice of T , OpenScholar still performs retrieval and generation largely separately. ReClaim [XWL⁺25] TODO

ScholarCopilot [WMN⁺25] is a RAG system developed for generating academic-style text with accurate citations. It is intended as a tool that assists the user with academic paper writing by providing iterative text generation with automated citation retrieval [WMN⁺25]. ScholarCopilot also enables user interaction during generation [WMN⁺25] by allowing the user to rewrite the generated text or forcefully trigger citation retrieval between the consecutive generation steps but this feature is beyond the scope of this thesis and is thus not utilized for the experiments.

3 Methods

TODO

3.1 ScholarCopilot

TODO

4 Experiments

TODO

5 Conclusion

TODO

Bibliography

- [AHS⁺24] Akari Asai, Jacqueline He, Rulin Shao, Weijia Shi, Amanpreet Singh, Joseph Chee Chang, Kyle Lo, Luca Soldaini, Sergey Feldman, Mike D’arcy, David Wadden, Matt Latzke, Minyang Tian, Pan Ji, Shengyan Liu, Hao Tong, Bohao Wu, Yanyu Xiong, Luke Zettlemoyer, Graham Neubig, Dan Weld, Doug Downey, Wen tau Yih, Pang Wei Koh, and Hannaneh Hajishirzi. Openscholar: Synthesizing scientific literature with retrieval-augmented lms, 2024.
- [SHC⁺24] Aviv Slobodkin, Eran Hirsch, Arie Cattan, Tal Schuster, and Ido Dagan. Attribute first, then generate: Locally-attributable grounded text generation, 2024.
- [WMN⁺25] Yubo Wang, Xueguang Ma, Ping Nie, Huaye Zeng, Zhiheng Lyu, Yuxuan Zhang, Benjamin Schneider, Yi Lu, Xiang Yue, and Wenhua Chen. ScholarCopilot: Training large language models for academic writing with accurate citations, 2025.
- [XWL⁺25] Sirui Xia, Xintao Wang, Jiaqing Liang, Yifei Zhang, Weikang Zhou, Jiaji Deng, Fei Yu, and Yanghua Xiao. Ground every sentence: Improving retrieval-augmented llms with interleaved reference-claim generation, 2025.